

part of deriving their equations—we just have to memorize them and use them appropriately.

The Inner Physicist

Using Galileo's and al-Khwarizmi's equations and a solve-for- x strategy, a sophisticated machine can catch a ball or intercept a missile. But it seems reasonable to assume that most baseball players don't start writing out equations as soon as they see a ball go into the air. Reliable observers have reported that professional baseball spring training programs consist of a great deal of time running around and playing, and considerably less time gathered around a whiteboard deriving the Navier-Stokes equations. Solving the mystery of where a ball will land doesn't provide a clear-cut answer to the outfielder problem—that is, how a *human* can instinctively know where a ball will land without plugging it into a computer program.

Or maybe it does. The glibbest possible solution to the outfielder problem is to assert that if computers are solving Galilean quadratics to determine where balls will land, then so are humans. We'll call this solution the *inner physicist theory*. According to this theory, the “wetware” of our brains is able to set up and solve quadratic equations, or else draw plots and extrapolate their lines, all far beneath the level of our consciousness. Each of us, in other words, has an “inner physicist” deep in our brains who can calculate exact solutions to difficult math problems in seconds and deliver the solutions to our muscles, which can then find their way to the ball, bringing our bodies and mitts along. Our subconscious might be able to do this even if we've never taken a physics class or solved for x .

The inner physicist theory is not without its proponents. Notably, the well-known mathematician Keith Devlin published a book in 2006 called *The Math Instinct: Why You're a Mathematical Genius (Along with Lobsters, Birds, Cats, and Dogs)*. The book's cover shows a dog jumping to catch a Frisbee, with arrows tracing the respective trajectory vectors of the Frisbee and the dog, implying that the dog is able to perform the intricate calculations that would be required to make those vectors meet.

The manifest ability of dogs to catch Frisbees and humans to catch baseballs seems to be a point in favor of the inner physicist theory. The subconscious is a mysterious and powerful thing, whose depths we have yet